THE INTERNAL-HOM EVALUATE MAP IS AN ISOMORPHISM UNDER PERFECTION HYPOTHESES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The internal hom of derived quasi-coherent objects with the stated hypotheses lands in D_QCoh, so the assertion is local on X. invoked at proof lines 2738-2739. Reduces cases (1) and (3) to affines.
- On affines the evaluate map is an isomorphism in the module setting under the corresponding boundedness hypotheses. invoked at proof lines 2739-2741. Treats the affine cases (1) and (3).
- On affines pseudo-coherence, perfectness, and finite tor dimension of objects agree with the corresponding module properties. invoked at proof lines 2742-2744. Performs the translation into algebra.
- Locally a perfect object is a finite complex of finite free modules, for which the evaluate map is visibly an isomorphism. used at proof lines 2745-2749. Treats case (2) without quasi-coherence of either side.

Source reference. Original lemma 0BYS: sources/stacks-chow/tags.txt, source lines 15025–15025. Cited at statement lines 2727-2728 as the map whose isomorphism is the claim; the proof of the isomorphism is inside the counted span.

Source reference. Original lemma 0A6H: sources/stacks-perfect/pinned-original-source.tex, source lines 2657–2657. Cited at proof lines 2738-2739 to reduce cases (1) and (3) to affines; the translations and conclusions are inside the counted span.

Source reference. Original lemma 0ATK: sources/stacks-chow/tags.txt, source lines 13618–13618. Cited at proof lines 2739-2741 for the affine cases (1) and (3); the reduction is inside the counted span.

Source reference. Original lemma 08E7: sources/stacks-perfect/pinned-original-source.tex, source lines 2440–2440. Cited at proof lines 2742-2743 in the translation into algebra; the reduction is inside the counted span.

Source reference. Original lemma 08EB: sources/stacks-perfect/pinned-original-source.tex, source lines 2636–2636. Cited at proof lines 2742-2743 in the translation into algebra; the reduction is inside the counted span.

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Source reference. Original lemma 08E9: sources/stacks-perfect/pinned-original-source.tex, source lines 2508–2508. Cited at proof lines 2743-2744 in the translation into algebra; the reduction is inside the counted span.

Lemma 0.2. *Let X be a scheme. Let K, L, M be objects of $D_{QCoh}(\mathcal{O}_X)$. The map*

$$K \otimes^{\mathbf{L}}_{\mathcal{O}_X} R\mathcal{H}om(M, L) \longrightarrow R\mathcal{H}om(M, K \otimes^{\mathbf{L}}_{\mathcal{O}_X} L)$$

of Cohomology, Lemma 0BYS is an isomorphism in the following cases

(1) *M perfect, or*
(2) *K is perfect, or*
(3) *M is pseudo-coherent, $L \in D^+(\mathcal{O}_X)$, and K has finite tor dimension.*

Proof. Lemma 0A6H reduces cases (1) and (3) to the affine case which is treated in More on Algebra, Lemma 0ATK. (You also have to use Lemmas 08E7, 08EB, and 08E9 to do the translation into algebra.) If K is perfect but no other assumptions are made, then we do not know that either side of the arrow is in $D_{QCoh}(\mathcal{O}_X)$ but the result is still true because we can work locally and reduce to the case that K is a finite complex of finite free modules in which case it is clear. □